

OIL INDIA LIMITED · HEALTH, SAFETY AND ENVIRONMENT

SENTRA

Safety · Intelligence · Compliance

System design flow and screen-by-screen prototype

SIF-precursor detection in near-miss and UA/UC reports - Smart India Hackathon, Problem Statement 26165.

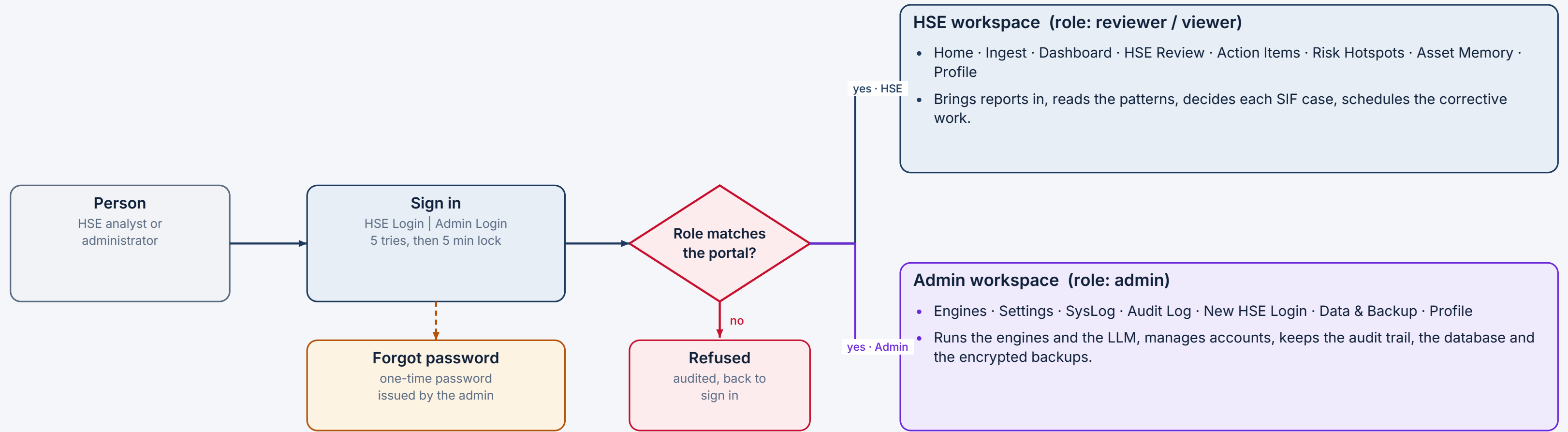
Version 2.0 · 27 September 2026

CONTENTS

- 1 User journey and roles
- 2 Report processing pipeline
- 3 System architecture
- 4 Navigation map (prototype)
- 5 Screens, one by one - click a screen on the map to jump
- 6 Decision logic, security and data

Who goes where

two roles, one front door

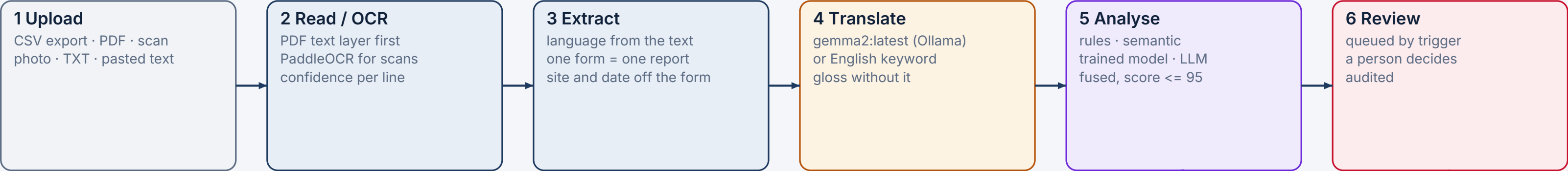


What every screen shares

Title row	OIL emblem · SENTRA · workspace badge · PS 26165 · gemma2:latest status (green = answering) · alerts bell · theme · the person and sign out
Audit	every sign-in, decision, export, setting and account change is written to a hash-chained audit log
Dates	one date format and time zone (Settings), shown with the zone - 27 Sep 2026, 14:06 IST
Permissions	each action checks the role (admin, reviewer, viewer); a refused action is audited

From a report to a decision

what happens to every document



Inside 5 · Analyse - the fused assessment

Rule engine	IOGP Life-Saving Rules, high-energy and barrier signatures (regex, no model)
Semantic encoder	sentence vectors against labelled prototypes
Trained model	learned from reviewed decisions (MLflow-tracked)
Local LLM	gemma2:latest reads the report; disagreement goes to a person
SIF rule	SIF potential = high-energy source AND a failed critical barrier
Risk score	0-95 (never 100); critical at 85 and above; P(SIF) <= 0.95

After the analysis

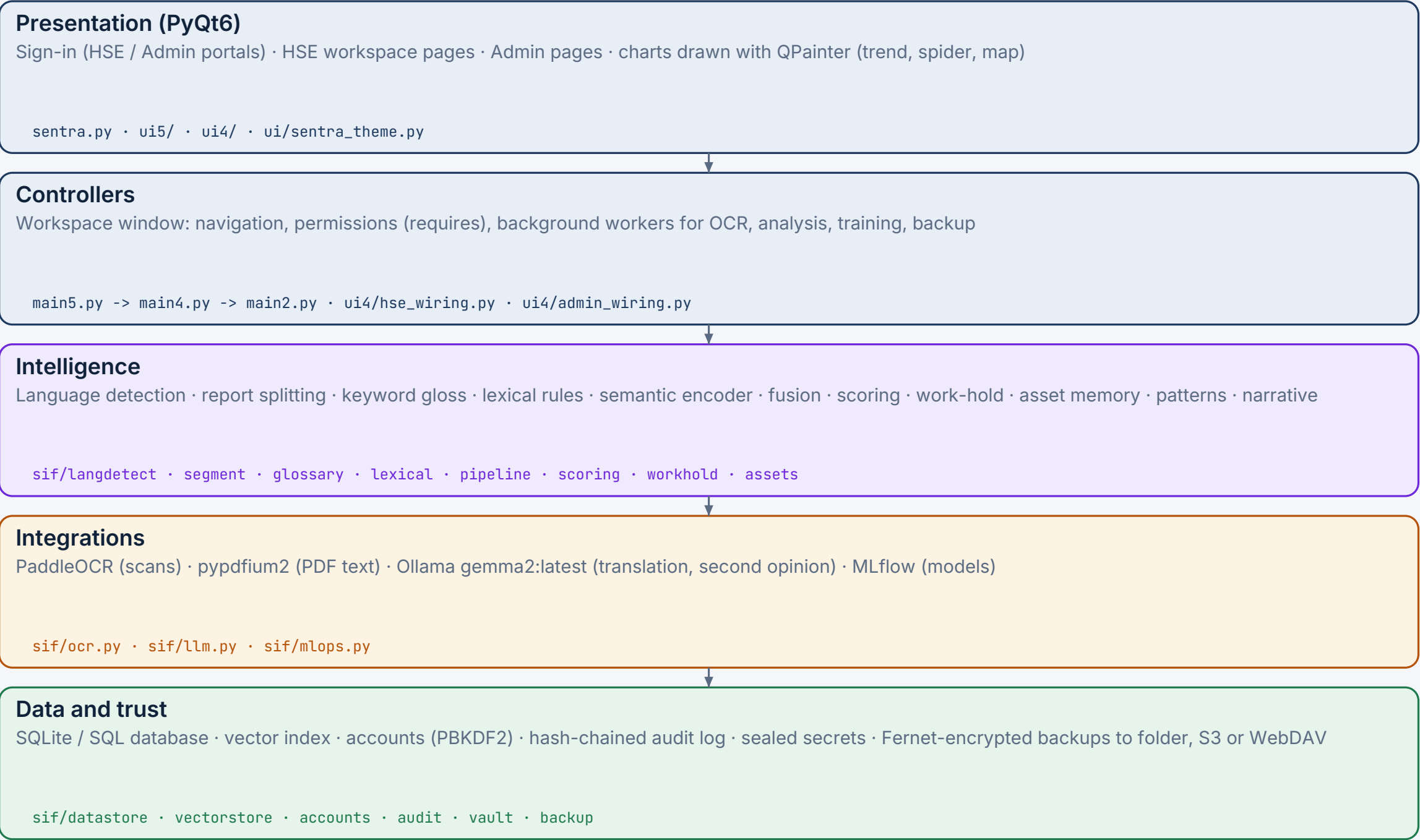
Work-hold	weighted evidence: hold at 12, review at 4 - routed to an HSE expert
Review queue	triggers: critical risk, energy without barrier, thin evidence, LLM disagreement
Decision	Confirm SIF · Not SIF · Need more information - under the person's name
Memory	each asset's history: recurring hazards, repeated control failures
Patterns	dashboard trend, spider chart, hotspots map, asset memory
Record	SQL database, vector index, hash-chained audit, encrypted backup

Languages

The language is read from each report's own letters: Hindi, Marathi, Assamese, Bengali, Tamil, Telugu, Kannada, Malayalam, Gujarati, Punjabi, Odia, Urdu, English. With Ollama the report is translated before analysis; without it the engines read an English keyword gloss of its safety terms (no LOTO, no gas test, expired permit, under the suspended load...). The original is always kept as the record.

Layers and where they live

desktop, offline-first



Runs on

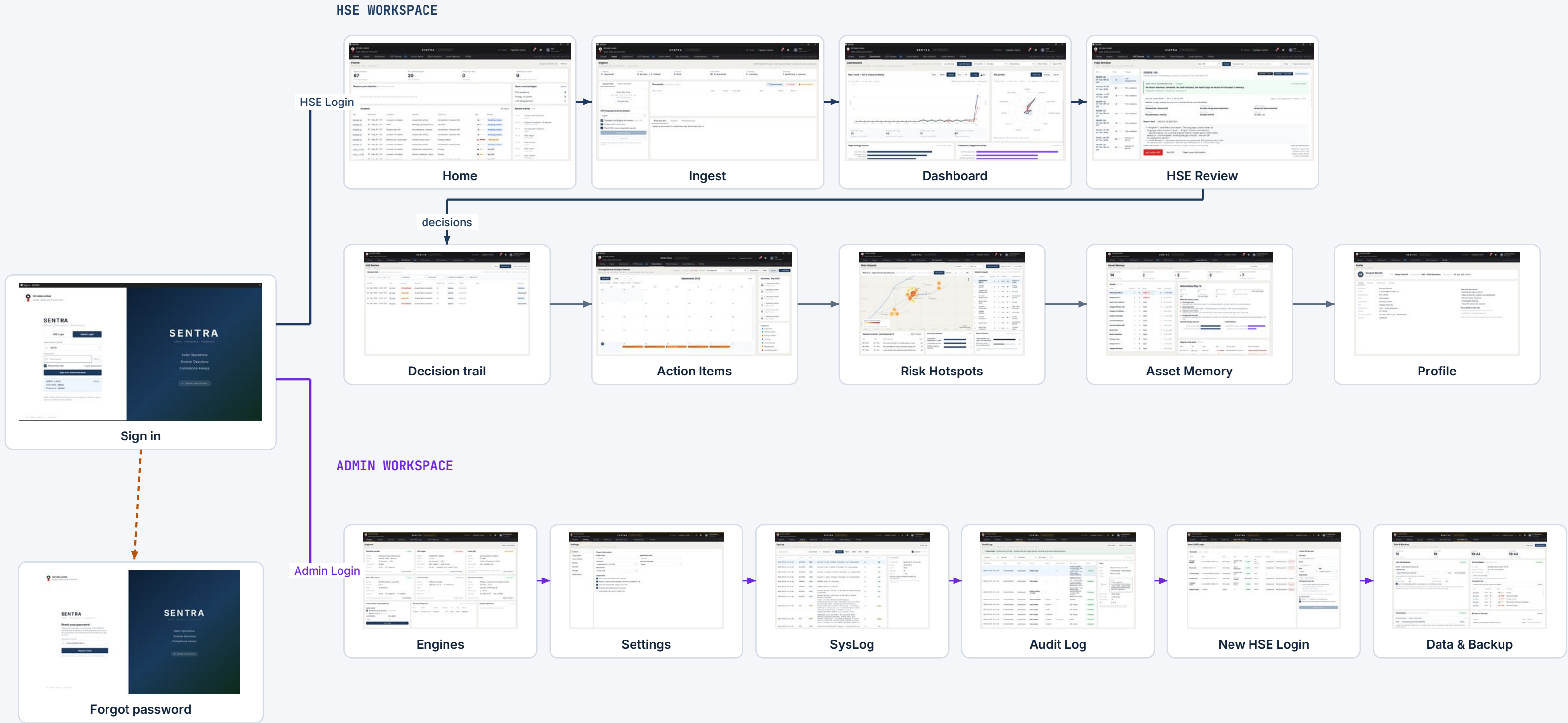
- Windows desktop - SENTRA.exe from the Inno Setup installer
- Data in %APPDATA%\SIF Insight Console, never in the program folder
- Works offline; Ollama and cloud backup are optional

Accounts and roles

- admin - Administration: engines, settings, accounts, audit, data
- reviewer (HSE Analyst) - ingest, analyse, decide, actions
- viewer - read only
- Built-in: admin / sih165, hse / sih2026165 - change before real use

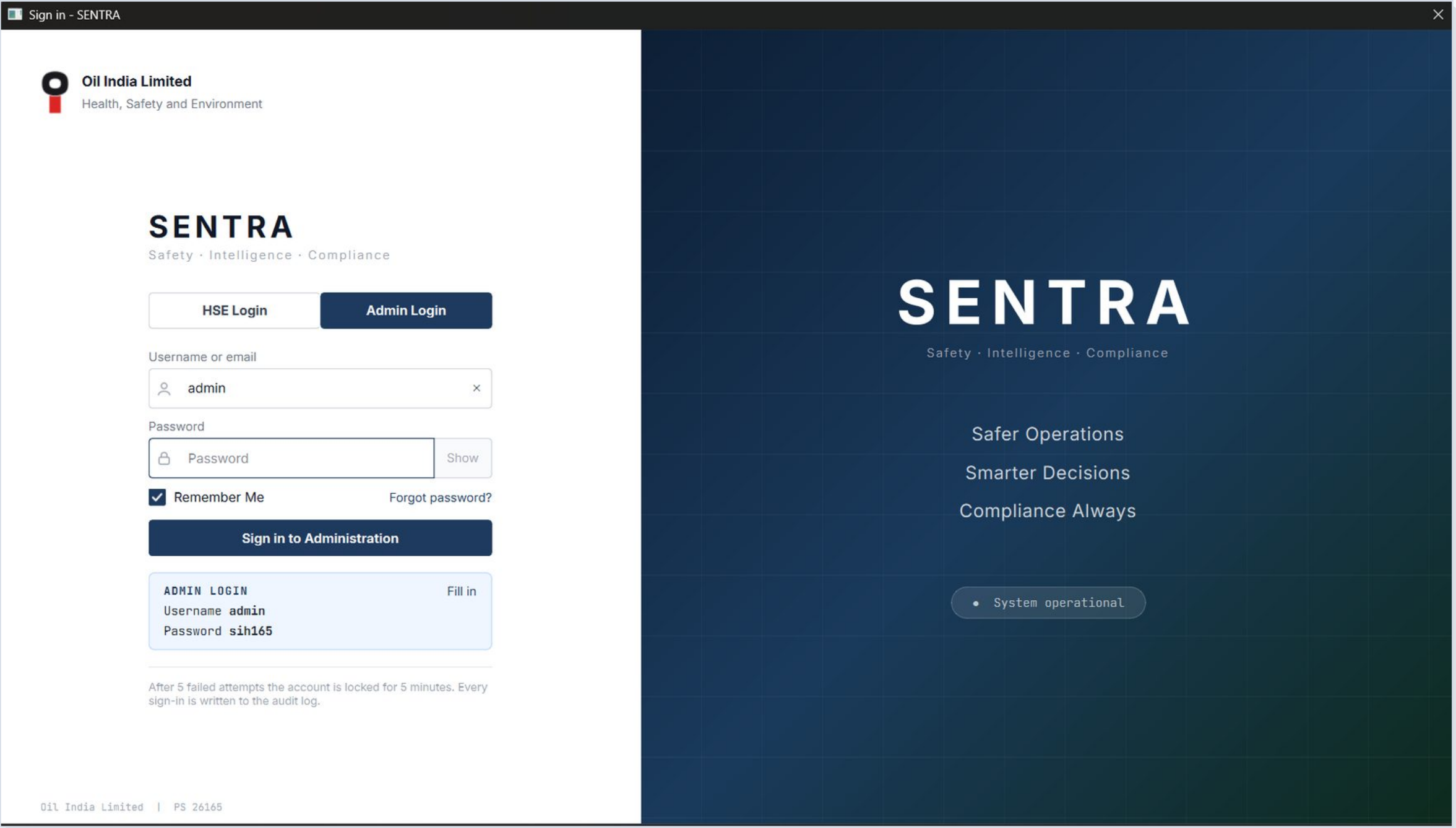
The prototype - click any screen to open it

arrows = the path a person takes



The tab bar reaches every page directly; the arrows show the usual order of work. Each thumbnail links to its screen page.

Sign in - HSE Login / Admin Login



PURPOSE

One door, two portals. The person picks HSE Login or Admin Login and signs in; SENTRA opens straight into the matching workspace.

THE PERSON

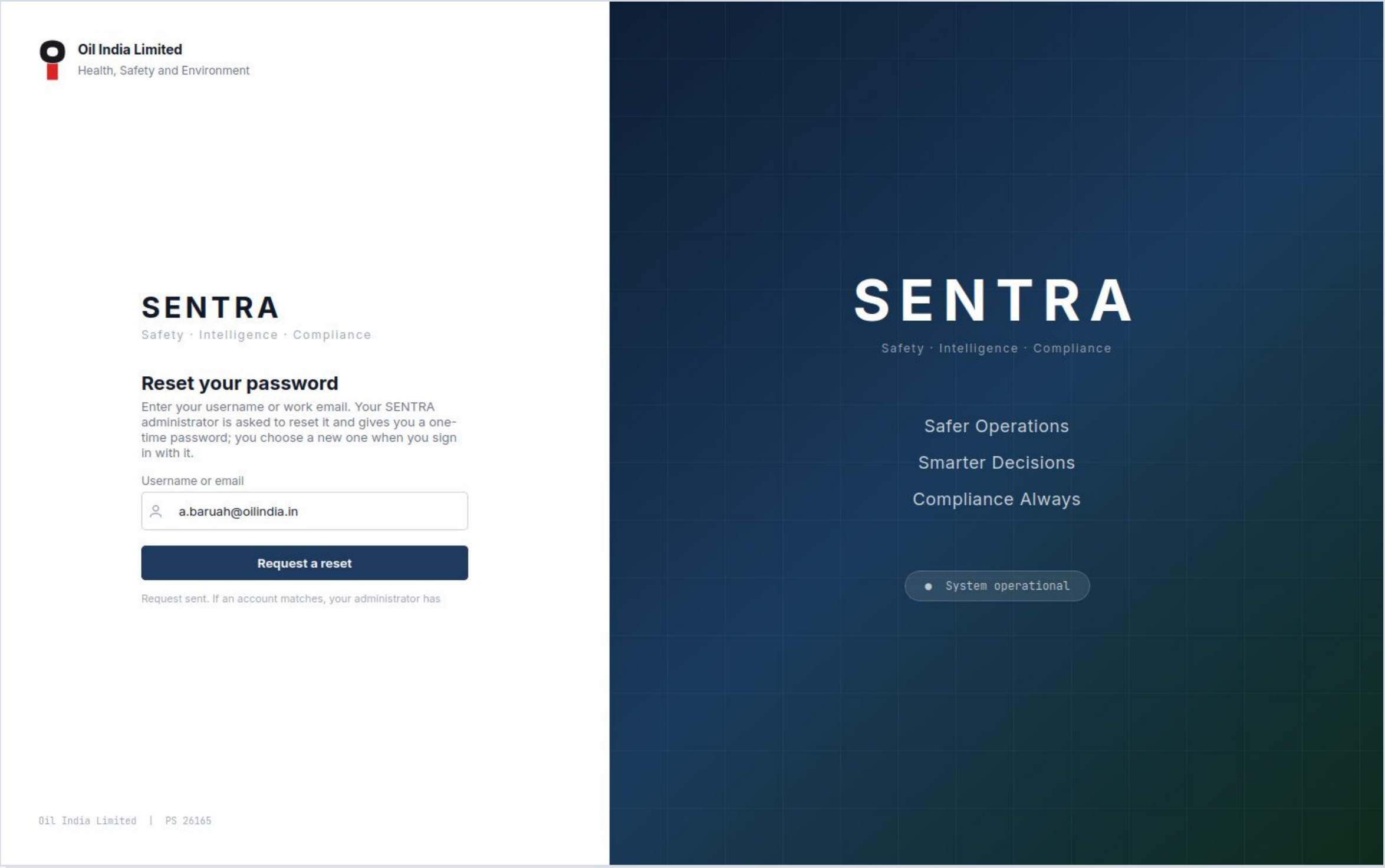
- Choose HSE Login or Admin Login
- Enter username (or email) and password
- Remember Me, Show password, Forgot password?

SENTRA

- Checks the account's role against the portal - an HSE account is refused at Admin Login
- Locks an account for 5 minutes after 5 failed attempts
- Writes every sign-in, success or failure, to the hash-chained audit log
- Shows the built-in credentials only while the default password is unchanged

NEXT ->
HSE Login -> Home. Admin Login -> Engines.

Forgot password



PURPOSE

A person who cannot sign in asks for a reset without seeing anyone else's account.

THE PERSON

- Enter username or work email
- Request a reset

SENTRA

- Asks the administrator to issue a one-time password (audited)
- The same answer whether or not the account exists
- On next sign-in with the one-time password, a new password is chosen

NEXT ->
Back to Sign in.

Home - what needs you now

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HSE WORKSPACE

PS 26165

gemma2:latest

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H

hse

HSE Analyst

Home

Ingest

Dashboard

HSE Review

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Action Items

Risk Hotspots

Asset Memory

Profile

Last analysis run 27 Sep 2026, 08:36 UTC

Home

Sun 27 Sep 2026 · 10:16 UTC

Updated 10:16:06 UTC

Refresh

Reports analysed

67

67 analysed today

SIF potential (engine)

29

43% of reports

Critical risk, open

0

none open

Awaiting your review

9

+8 needs info · 0 over 48 h

Requires your attention

worst first, then oldest

Nothing needs you right now. New reports that do will appear here, worst first.

Open cases by trigger

Queue

Thin evidence

6

Energy, no barrier

2

LLM disagreement

1

Recent incidents

All reports

Ref	Received	Location	Activity	IOGP rule	Risk	Review
README-06	27 Sep...15 UTC	Location not stated	Unspecified activity	Unclassified / General HSE	0	Awaiting review
README-05	27 Sep...14 UTC	Hindi	Welding / grinding (hot w...	Hot Work	21	Awaiting review
README-04	27 Sep...14 UTC	Duliajan field HQ	Housekeeping / cleaning	Unclassified / General HSE	0	Awaiting review
README-03	27 Sep...14 UTC	Location not stated	Unspecified activity	Unclassified / General HSE	0	Awaiting review
README-02	27 Sep...14 UTC	Warehouse / stores yard	Confined space entry	Energy Isolation	95	? Needs info
README-01	27 Sep...14 UTC	Location not stated	Unspecified activity	Unclassified / General HSE	0	Awaiting review
urdu_mrt-03	27 Sep...24 UTC	Location not stated	Component replacement ...	Driving	44	Not SIF
urdu_mrt-02	27 Sep...24 UTC	Location not stated	Vehicle movement / trans...	Driving	58	Not SIF
urdu_mrt-01	27 Sep...23 UTC	Location not stated	Unspecified activity	Unclassified / General HSE	0	Not SIF

Recent activity

today

10:16

Trained model attached

System

08:36

9 report(s) analysed · 18 queued

Analysis run by you

08:35

You read after_install.txt

Ingest

08:35

Files staged

Functionality

08:35

Ollama check

System

08:35

Data loaded

Functionality

08:34

Sign-in failed

Functionality

PURPOSE

The HSE analyst's start page: headline figures, the cases that need a person worst-first, open cases by trigger, recent incidents and recent activity.

THE PERSON

- Read the four figures: analysed, SIF potential, critical open, awaiting review
- Open a case from Requires your attention or Recent incidents
- Refresh; open the bell for alerts

SENTRA

- Orders the attention list by risk, then by how long it has waited
- Counts open cases per review trigger (thin evidence, energy without barrier, LLM disagreement...)
- Streams the day's activity from the audit log

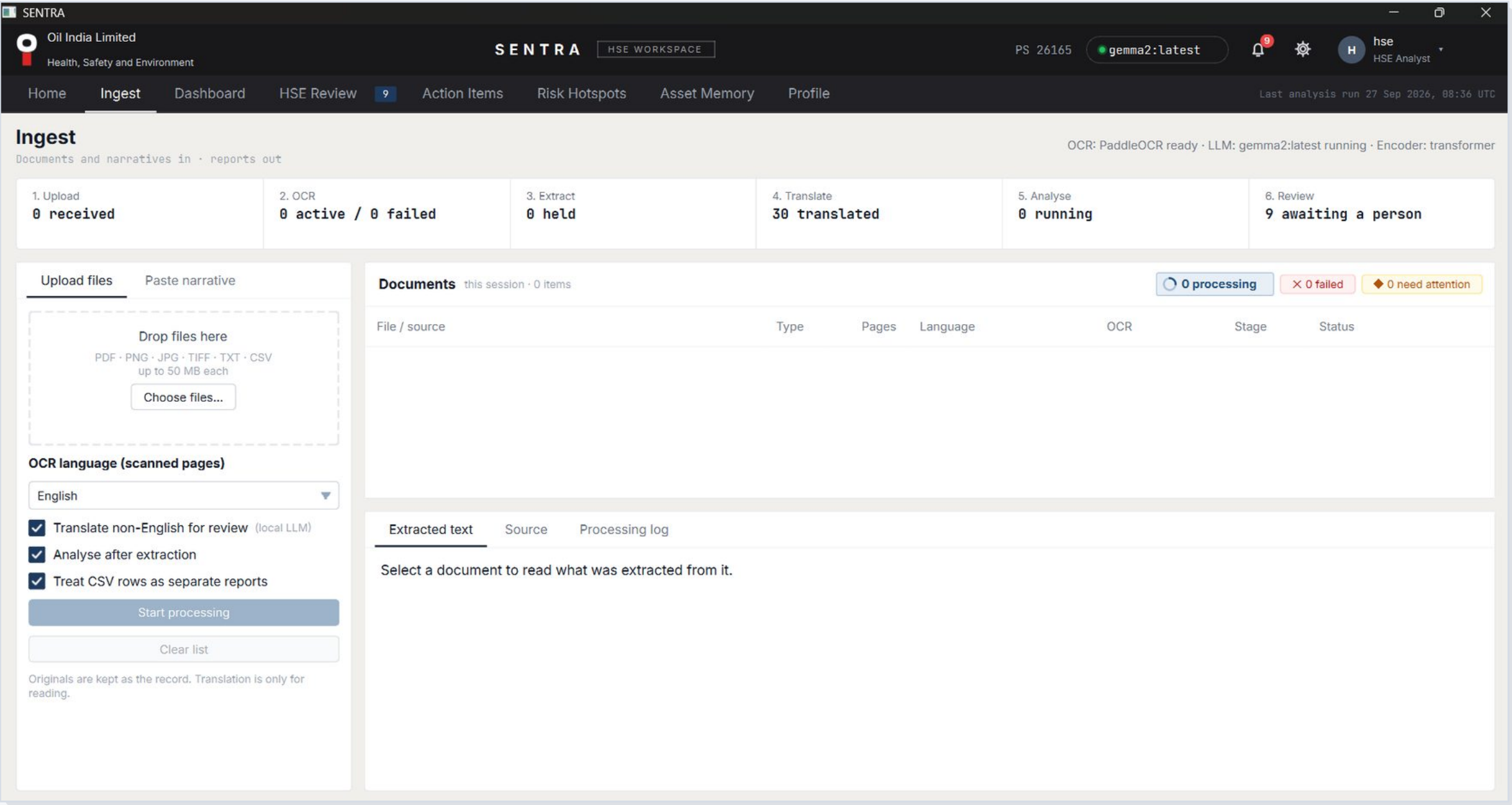
NEXT ->

Ingest to bring reports in, or HSE Review to decide cases.

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Ingest - documents and narratives in, reports out



PURPOSE

Every way a report arrives: CSV exports, PDFs, scans and photographs, text files, or a pasted narrative - through a six-stage pipeline with a log per document.

THE PERSON

- Drop or choose files (PDF, PNG, JPG, TIFF, TXT, CSV) or paste a narrative
- Set the OCR language for scanned pages only
- Start processing; Cancel or Clear list

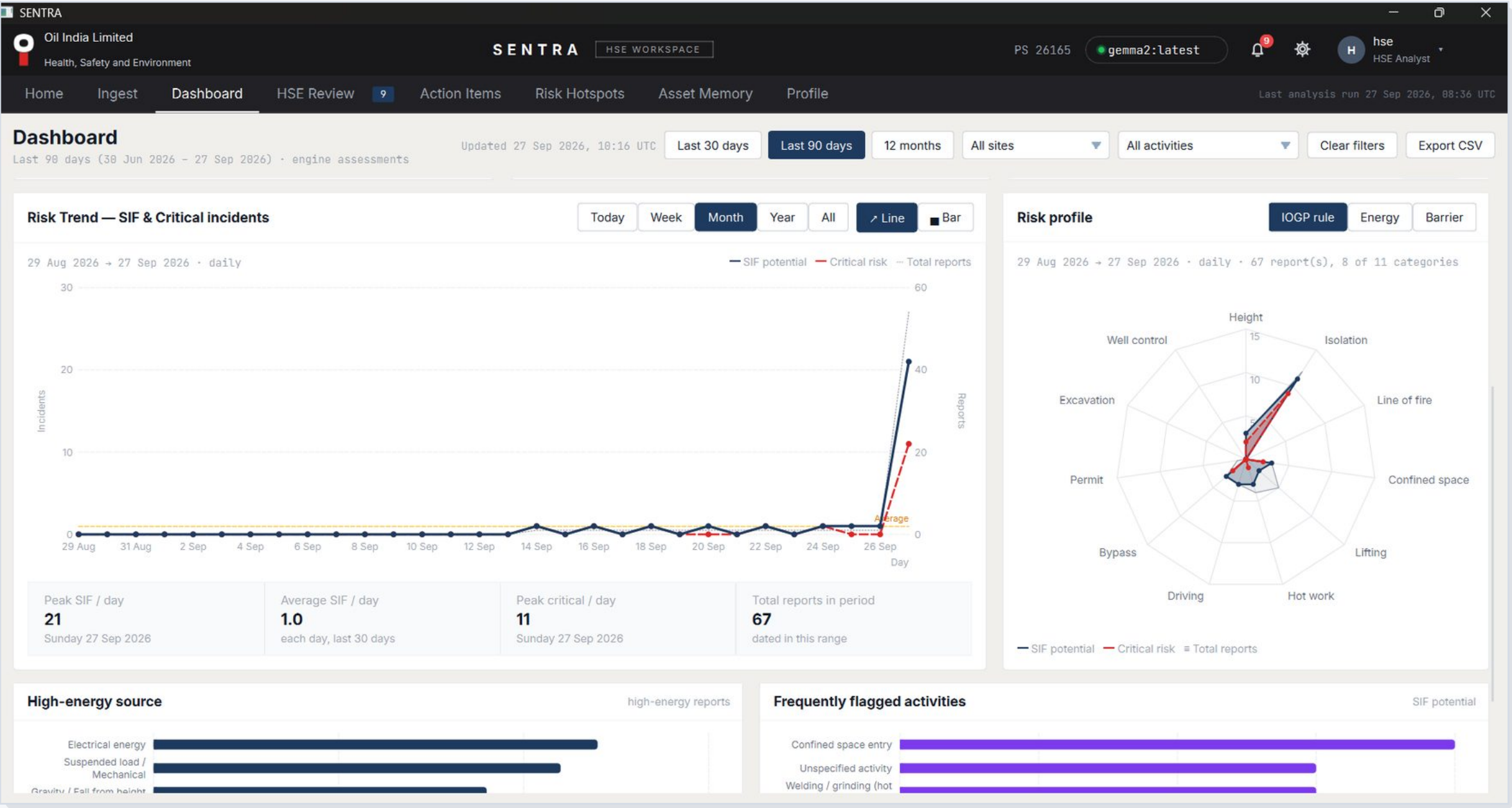
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- 1 Upload 2 OCR (PaddleOCR) 3 Extract 4 Translate 5 Analyse 6 Review
- Detects each report's language from its text; keeps a report form as one report
- Translates with gemma2:latest, or reads an English keyword gloss without it
- Keeps the original as the record; reads site and date off the form

NEXT ->

Dashboard for the picture, HSE Review for the cases.

Dashboard - risk trend and risk profile



PURPOSE

Where the precursor patterns show: the risk trend up to now and a spider chart of the same reports, then high-energy sources and flagged activities.

THE PERSON

- Pick the period, site and activity; Clear filters; Export CSV
- Trend range: Today / Week / Month / Year / All; Line or Bar
- Spider chart by IOGP rule, Energy or Barrier; hover for figures

SENTRA

- Buckets reports by hour, day or month, ending now
- Counts SIF potential, critical (risk >= 85) and all reports
- Peak, average and total per range beneath the chart

NEXT ->

HSE Review to act on what the chart shows.

HSE Review - the case and the decision

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PS 26165

gemma2:latest

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hse

HSE Analyst

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Ingest

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HSE Review

Action Items

Risk Hotspots

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Last analysis run 27 Sep 2026, 08:36 UTC

HSE Review

9 open · decisions are recorded under hse

Ref	Risk	Trigger
README-04 27 Sep 08:14 UTC	0	LLM disagreement
kannad...fi-01 27 Sep 2026	10	Thin evidence
assame...w-01 17 Sep 2026	9	Thin evidence
README-01 27 Sep 08:14 UTC	0	Thin evidence
README-03 27 Sep 08:14 UTC	0	Thin evidence
README-06 27 Sep 08:15 UTC	0	Thin evidence
bengal...st-01 21 Sep 2026	0	Thin evidence
hindi...nt-01 19 Sep 2026	24	Energy, no barrier
README-05 27 Sep 08:14 UTC	21	Energy, no barrier

README-04

Duliajan field HQ · Housekeeping / cleaning · analysed 27 Sep 2026, 08:14 UTC

ENGINE · Low 0

ENGINE · not SIF

Awaiting review

WORK-HOLD RECOMMENDATION

Continue

no routing needed

No factor reached a threshold. No hold indicated; the report stays on record for the asset's memory.

Weighted evidence 0 · hold at 12, review at 4

ENGINE ASSESSMENT — NOT A DECISION

Trigger: LLM disagreement · waiting 2 h 2 m

Neither a high-energy source nor a barrier failure was identified.

IOGP rule

Unclassified / General HSE

Activity

Housekeeping / cleaning

Energy source

No high-energy source identified

Location

Duliajan field HQ

Failed barrier

No barrier failure identified

Source

README .md

Report text

ENGLISH AS WRITTEN

* **Ingest** - each file is one report. The Language column names its language; after analysis it reads `→ English` (Ollama translated it), `· keyword gloss` (no LLM: the engines read an English gloss of the safety terms) or `· not translated` (nothing the gloss knows - the non-SIF housekeeping reports).

* **HSE Review** - the seven precursors are queued as SIF potential, each with its site (Duliajan, Nabarkativa, Moran) read off the form. A non-English case

HUMAN DECISION

recorded under hse (HSE Analyst) · written to the audit log

Confirm SIF

Not SIF

? Need more information

Undo the last decision

Read the report and evidence first. The engine result above is not a decision.

PURPOSE

The case list beside the case: work-hold recommendation, the engine's assessment (not a decision), the report text, the asset's memory, evidence - and three keys to decide.

THE PERSON

- Pick a case; read the recommendation, facts and report
- Show the original / translation / keyword gloss
- Confirm SIF, Not SIF or Need more information; Undo the last decision

SENTRA

- Work-hold: weighted evidence - hold at 12, review at 4
- Records the decision under the signed-in person, in the audit log and the SQL store
- A rejection releases a hold; the decision trail keeps every version

NEXT ->

Decision trail, or Action Items to raise a corrective action.

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HSE Review - Decision trail

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Anupam Baruah

HSE Analyst

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Last analysis run 26 Sep 2026, 14:49 IST

HSE Review

16 open · decisions are recorded under Anupam Baruah

Cases

Decision trail

Export decision trail

Decision trail

every decision recorded, newest first

4 of 4 shown · 1 confirmed, 0 not SIF, 1 need info · 0 overturned the engine

Search ref, reviewer, trigger, note

All reviewers

All decisions

Standing and superse

Clear filters

Decided	Ref	Decision	Reviewer	Engine said	Overturns	Trigger	Note	Standing
26 Sep 2026, 14:49 IST	NM-2601	SIF confirmed	Anupam Baruah (a.baruah)	SIF	Agrees	Critical risk		Standing
26 Sep 2026, 14:49 IST	NM-2601	Needs info	Anupam Baruah (a.baruah)	SIF	Agrees	Critical risk		Superseded
26 Sep 2026, 14:49 IST	NM-2602	Needs info	Anupam Baruah (a.baruah)	SIF	Agrees	Critical risk		Standing
26 Sep 2026, 14:49 IST	NM-2601	SIF confirmed	Anupam Baruah (a.baruah)	SIF	Agrees	Critical risk		Superseded

PURPOSE

Every decision recorded, standing or superseded, filterable by reviewer and decision, exportable.

THE PERSON

- Switch Cases -> Decision trail
- Filter by reviewer or decision; search
- Export decision trail (CSV)

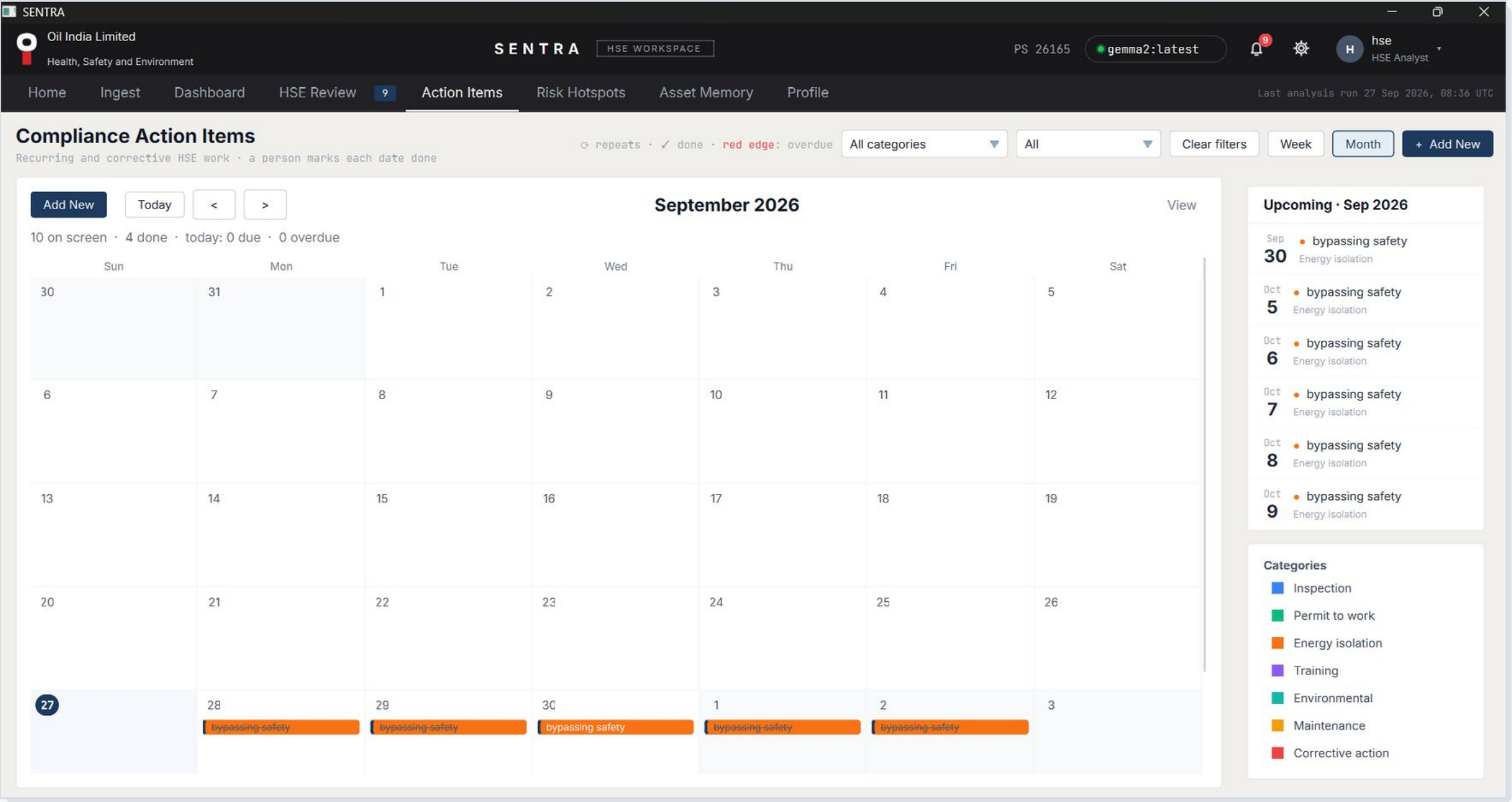
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- Shows who decided what, when, and what it replaced
- The export is audited like every other export

NEXT ->

Back to Cases.

Action Items - the compliance calendar



PURPOSE

Recurring and corrective HSE work by category, month or week, with what is due next.

THE PERSON

- Add New (one-off or repeating); Week / Month; Today, < >
- Mark a date done; filter by category and status; Clear filters

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- Colours by category: inspection, permit, energy isolation, training...
- Red edge for overdue; the upcoming list on the right
- Every add, done, reopen and delete is audited

NEXT ->
Risk Hotspots.

Risk Hotspots - the interactive map

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PS 26165 gemma2:latest

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AB Anupam Baruah
HSE Analyst

Home Ingest Dashboard HSE Review **16** Action Items Risk Hotspots Asset Memory Profile
Last analysis run 26 Sep 2026, 14:41 IST

Risk Hotspots

Repeats ranked by SIF-precursor density, not by report count

All reports ▾ Every site ▾
 Rank by density Rank by count Clear filters

Risk map — Upper Assam operating area

hover for figures · click to choose · scroll to zoom · drag to pan

SIF density
Reports
+
-
Fit

Colour: SIF-precursor density
0% 100%

Size: number of reports · positions approximate

Ranked hotspots

density = lower bound of SIF share (Wilson 95%)

#	Location	Reports	SIF	Density	Dominant rule
1	Naharkatiya Rig-12	3	3	44%	Working at height
2	Duliajan OCS-4	2	2	34%	Energy isolation
3	Duliajan LPG Bottling Plant	1	1	21%	Hot work
4	Kumchai Pipeline RoW	1	1	21%	Excavation & ground ...
5	Moran Field Road	1	1	21%	Driving
6	Kumchai Drilling Site	1	1	21%	Well control & process ...
7	Duliajan Effluent ...	1	1	21%	Line of fire
8	Duliajan Substation	1	1	21%	Energy isolation
9	Moran Field	1	1	21%	Excavation & ground ...
10	Moran Gas Compress...	2	1	10%	Confined space

Repeated incidents · Naharkatiya Rig-12

[Open in Review](#)

Ref	Date	What happened	Risk
NM-2616	23 Jan	Third report from Rig-12: scaffold platform erecte...	90
NM-2606	12 Jan	The crane lifted a 4 tonne mud pump module over...	75
NM-2602	06 Jan	A crew member was replacing a light fitting on the...	95

Activity breakdown

3 reports

Component replacement / repair	1
Unspecified activity	1
Drilling / tripping operation	1

Barrier failures

repeat = same barrier ≥ 2

Fall protection not used / not anchored	2
Exclusion zone / barricading absent	1

Dominant: Fall protection not used / not anchored under Working at height.

The Upper Assam operating area: each site at its locality, sized by reports and coloured by SIF-precursor density.

- Hover a site for its figures; click to choose it
- Scroll to zoom, drag to pan
- Read the ranked table and the chosen site's incidents

- Ranks by density (Wilson lower bound), not volume
- Clusters by location, activity, rule-at-location and repeated barrier failure

Asset Memory

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Asset Memory

Profile

Last analysis run 28:13

Asset Memory

Each asset's incidents, near misses, hazards, control failures and corrective actions

Assets on record

14

18 report(s) placed

Recurring hazards

2

assets, last 90 days

Repeated control failures

2

assets, last 90 days

Emerging SIF precursors

2

assets

Work holds recommended

7

awaiting an HSE decision

Assets

worst first

Asset	Reports	SIF	History	Holds	Last report
Naharkatiya Rig-12	3	3	4 high · 4	2	23 Jan 2026
Duliajan OCS-4	2	2	3 high · 4	1	14 Jan 2026
Moran Gas Compress...	2	1	Quiet	1	19 Jan 2026
Duliajan Effluent Treat...	1	1	Quiet	1	22 Jan 2026
Duliajan LPG Bottling ...	1	1	Quiet	0	9 Jan 2026
Duliajan Substation	1	1	Quiet	1	24 Jan 2026
Kumchai Drilling Site	1	1	Quiet	0	21 Jan 2026
Kumchai Pipeline RoW	1	1	Quiet	0	11 Jan 2026
Moran Field	1	1	Quiet	1	25 Jan 2026
Moran Field Road	1	1	Quiet	0	15 Jan 2026
Duliajan Camp	1	0	Quiet	0	17 Jan 2026
Duliajan OCS-2	1	0	Quiet	0	20 Jan 2026
Duliajan Workshop	1	0	Quiet	0	16 Jan 2026

Naharkatiya Rig-12

Reports

3

Incidents

0

Near misses

3

Hazard observations

0

SIF precursors

3 (0 confirmed)

Peak risk

100

Last report

23 Jan 2026

Open actions

0

Equipment named

-

What the history says

▲ Recurring hazard

Suspended load / Mechanical: report 2 at Naharkatiya Rig-12 in 90 days - earlier ones had fatal potential

▲ Recurring hazard

Gravity / Fall from height: report 2 at Naharkatiya Rig-12 in 90 days - earlier ones had fatal potential

▲ Repeated control failure

Fall protection not used / not anchored: failed 1 time(s) before at Naharkatiya Rig-12, first 6 Jan 2026

▲ Emerging SIF precursor

Gravity / Fall from height with "Fall protection not used / not anchored": 2 reports with fatal potential at Naharkatiya Rig-12 in 90 days

Hazards (energy sources)

Gravity / Fall from height

Suspended load / Mechanical

Control failures

Fall protection not used / not anchored

Exclusion zone / barricading absent

Reports at this asset

newest first

Date	Ref	Kind	Risk	Control failure	Recommendation
23 Jan 26	NM-2616	Near miss	94	Fall protection not used / n...	Work-Hold Recommended
12 Jan 26	NM-2604	Near miss	70	Exclusion zone / barricading a...	Work-Hold Recommended

PURPOSE

Each asset's incidents, near misses, hazards, control failures and corrective actions - worst first - and what its history says.

THE PERSON

- Find an asset or pick one from the list (reports, SIF, history, holds, last report)
- Read the figures: reports, incidents, near misses, hazard observations, peak risk
- Open a report from the asset's history; Clear

SENTRA

- Counts assets on record, recurring hazards, repeated control failures, emerging SIF precursors and work holds recommended
- Explains the history: recurring hazard, repeated control failure, emerging precursor, confirmed precursor on record
- Feeds the work-hold recommendation with the asset's past

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Profile.

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Profile

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Profile

Last analysis run 19:43

Profile

Your account on SENTRA

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Anupam Baruah

HSE Analyst - Oil India Limited

SiteDuliajan Field HQ

DepartmentHSE — Field Operations

Last sign-in25 Sep 2026 19:43

Profile

Security

Preferences

Activity

Full name

Anupam Baruah

Email

a.baruah@oilindia.in

Employee no.

0IL-48213

Role

HSE Analyst

Organisation

Oil India Limited

Site

Duliajan Field HQ

Department

HSE — Field Operations

Project

PS 26165

Account created

25 Sep 2026 by D. Manikandan

Session

1ca91afc

What this role can do

• Upload and ingest reports

• Read all reports, evidence and dashboards

• Record review decisions

• Investigate hotspots

• Keep the action-item calendar

Not available to this role

• Engines, settings, system log, audit log

• Creating or changing accounts

Name, role and site are managed by your administrator.

PURPOSE

The person's own account, and exactly what their role can and cannot do.

THE PERSON

- Tabs: Profile, Security (change password), Preferences, Activity
- Menu under the name: My profile, Preferences, Sign out

SENTRA

- Shows role, organisation, project, when and by whom the account was created, the session
- Lists what this role can do, and what is not available to it
- Name, role and site are managed by the administrator

NEXT ->

Sign out -> Sign in.

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Administration - Engines

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ADMINISTRATION

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gemma2:latest

DM

D. Manikandan

Administrator

Engines

Settings

SysLog

Audit Log

New HSE Login

Data & Backup

Profile

vm

v2.0.0

Engines

host vm

Check all connections

Semantic encoder

✓ Ready

Model

Hashing encoder (lexical only)

Version

lexical-hash (512-d)

Host

In-process · CPU

Connection

Loaded

Last run

19:43 · 18 reports

Offline fallback: hashing encoder (lexical only)

Switch encoder

OCR engine

✕ Unavailable

Model

PaddleOCR · English

Version

3.7.0

Host

In-process · CPU

Connection

Not usable - see Verify

Last run

19:44 · scanned_uauc_report.png

Confidence floor 0.70 · 0 failure(s) this session

Verify OCR models

Local LLM

◆ Not reached

Model

gemma2:latest via Ollama

Version

Ollama

Host

http://localhost:11434

Connection

not checked yet

Last run

-

Used to translate non-English reports for reading and as a fourth opinion

Test connection

Risk / SIF analysis

✓ Ready

Model

SENTRA pipeline · lexical KB

Version

2.0.0

Host

In-process

Connection

—

Last run

19:43 · 18 reports · 17 queued

Compare the engine with reviewers' decisions.

Run evaluation

Learned model

Not trained

Model

XGBoost classifier

Version

xgboost 3.2.0 · 46 features

Host

-

Connection

—

Last run

-

Disagreements queued now: 0

Attach model

Experiment tracking

✓ Connected

Model

MLflow · experiment sif-insight-console

Version

mlflow 3.16.0

Host

sqlite:///mlflow.db

Connection

1 run(s)

Last run

10 Sep 15:37 · run 1c353a

Model files: models

Open run history

Train learned model (XGBoost)

Label source

Reviewed human decisions

1 available

Pipeline verdicts

(distillation)

n_estimators

220

max_depth

4

Train model

Recent training runs

MLflow · sif-insight-console

Run	Started	Labels	Samples	F1	AUC	Status
1c353a	10 Sep 15:37	reviewed...	56	1.000	1.000	Finished

Feature importance

run 1c353a

Train a model to see what it leans on.

NEXT

->

Settings.

PURPOSE

Every engine's model, version, host, connection and last run - one card each - and model training.

THE PERSON

- Semantic encoder: Switch encoder · OCR engine: Verify OCR models
- Local LLM (gemma2:latest via Ollama): Test connection · Check all connections
- Risk / SIF analysis: Run evaluation · Learned model (XGBoost): Detach model
- Train the learned model from reviewed decisions or pipeline verdicts; open run history

SENTRA

- Reports each engine Ready / Not reached / Models not fetched
- Tracks training runs in MLflow: labels, samples, F1, AUC, status, feature importance
- Queues engine-model disagreements for a person

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Administration - Settings

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SENTRA

ADMINISTRATION

PS 26165

gemma2:latest

DM

D. Manikandan
Administrator

EnginesSettingsSysLogAudit LogNew HSE LoginData & BackupProfile

Settings

Changes are written to the audit log with the value they replaced

General

Organization

Users & Roles

Models

Security

Storage

Notifications

Project information

Project code

PS 26165

Application name

SENTRA

Timezone

Asia/Kolkata (IST, UTC+5:30)

Interface language

English

Date format

25 Sep 2026

Application

☒ Auto-refresh HSE pages every 5 minutes

☒ Require a reason when a reviewer overturns an engine SIF call

☒ Allow file uploads (PDF, images, TXT, CSV)

☒ Translate non-English reports for review

☐ Show engine risk scores to Viewer role

PURPOSE

General, Organization, Users & Roles, Models, Security, Storage and Notifications.

THE PERSON

- Project code, application name, time zone, interface language, date format
- Auto-refresh, require a reason to overturn an engine SIF call, allow uploads, translate non-English reports, show risk scores to viewers

SENTRA

- Writes every change to the audit log with the value it replaced

NEXT ->

SysLog.

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Administration - SysLog

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vm · v2.0.0

SysLog

System events written by SENTRA services. No human actions appear here – see Audit Log.

Export .log

Search events

Last 24 hours

All services

All levels

ERROR

WARN

INFO

DEBUG

☒ Live tail 15 of 15

Timestamp	Service	Level	Event	Machine	Status
2026-09-25 19:44:02	ACCOUNTS	INFO	Account k.bora created (reviewer) by d.manikandan	vm	✓ OK
2026-09-25 19:44:02	ACCOUNTS	INFO	Account b.das created (reviewer) by d.manikandan	vm	✓ OK
2026-09-25 19:44:02	ACCOUNTS	INFO	Account p.saikia created (reviewer) by d.manikandan	vm	✓ OK
2026-09-25 19:44:02	ACCOUNTS	INFO	Account r.gogoi created (reviewer) by d.manikandan	vm	✓ OK
2026-09-25 19:44:02	CONSOLE	INFO	SENTRA (build 2) starting	vm	✓ OK
2026-09-25 19:44:02	CONSOLE	INFO	SENTRA (build 2) closing	vm	✓ OK
2026-09-25 19:44:00	REVIEW	INFO	Review decision unclear on NM-2602 by Anupam Baruah (a.baruah)	vm	✓ OK
2026-09-25 19:44:00	REVIEW	INFO	Review decision confirmed on NM-2601 by Anupam Baruah (a.baruah)	vm	✓ OK
2026-09-25 19:44:00	OCR	WARN	Could not read /home/user/SIF/samples/scanned_uavc_report.png: PaddleOCR could not start: No available model hosting platforms detected. Please check your network connection.. Its models download on first use, so an offline machine needs them pre-fetched into ~/.paddlex (or set PADDLE_PDX_MODEL_SOURCE to a reachable host).	vm	⚠ Warn
2026-09-25 19:44:00	OCR	WARN	PaddleOCR could not start: No available model hosting platforms detected. Please check your network connection.. Its models download on first use, so an offline machine needs them pre-fetched into ~/.paddlex (or set PADDLE_PDX_MODEL_SOURCE to ...	vm	⚠ Warn
2026-09-25 19:43:57	OCR	INFO	Initialising PaddleOCR (lang=en, orientation=True)	vm	✓ OK

Event detail

Timestamp

2026-09-25 19:44:02

Service

ACCOUNTS

Level

INFO

Machine

vm

Status

✓ OK

Account k.bora created (reviewer) by d.manikandan

Written by sif.accounts. Kept in /home/user/SIF/logs/sif_console.log.

PURPOSE

System events written by SENTRA's services - console, updater, training, review, OCR. No human actions here (see Audit Log).

THE PERSON

- Search; time range; service; level: ERROR, WARN, INFO, DEBUG; Clear filters
- Live tail; pick an event for its detail; Export .log

SENTRA

- Timestamp, service, level, event, machine and status for every entry
- Keeps the log file on disk (logs\sif_console.log)

NEXT ->
Audit Log.

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v2.0.0

Audit Log

Human actions only · append-only · each entry carries the hash of the one before

Verify chain

Export CSV for auditor

✓ Chain intact. 21 entries since 25 Sep · checked when this page opened · stored at /tmp/tmpuhzv1pyy/audit.jsonl

All users

All roles

All actions

Last 7 days

18 entries

Timestamp	User	Role	Action	Object	Previous value	New value	Result
2026-09-25 19:44:03	d.manikandan	Administrator	Backup made	—	—	archive: SENTRA-vm-20260925-1944... · bytes: 31492 · target: Folder · /tmp/tmpuhzv1pyy/OneDrive/SENTRA · ...	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	Data loaded	—	—	database: SQLite · /tmp/tmpuhzv1pyy/s.db · detail: pushed 0 new, 0 changed · pulled 18 report(s), 1 decision(s), 0 action...	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	Backup settings changed	—	—	target: Folder · /tmp/tmpuhzv1pyy/OneDrive/SENTRA · schedule: daily · keep: 14 · passphrase_change...	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	Account disabled	k.bora	Active	Disabled	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	User created	k.bora	—	HSE Analyst	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	User created	b.das	—	HSE Analyst	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	User created	p.saikia	—	HSE Analyst	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	Role changed	r.gogoi	HSE Analyst	Viewer	✓ Success
2026-09-25 19:44:02	d.manikandan	Administrator	User created	r.gogoi	—	HSE Analyst	✓ Success

Entry

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When

2026-09-25 19:44:03

Who

d.manikandan · Administrator

Action

Backup made

Object

—

Previous

—

New

archive: SENTRA-vm-20260925-194403.sentra · bytes: 31492 · target: Folder · /tmp/tmpuhzv1pyy/OneDrive/SENTRA · reports: 18

Entry hash

c031...3b4d

Prev. hash

edf8...6538

Workstation

vm

Build

2.0.0

Entries cannot be edited or deleted from SENTRA. A correction is a new entry that references this one.

PURPOSE

The hash-chained audit trail of human actions: sign-ins, decisions, exports, settings, accounts.

THE PERSON

- Filter by person, category and date; export; verify the chain

SENTRA

- Each entry carries the hash of the one before it - an edit breaks the chain

NEXT ->
New HSE Login.

Administration - New HSE Login

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New HSE Login

Create and manage HSE accounts · every change is written to the audit log

Accounts5 active · 1 disabled

Filter by name or email

Name	Email	Role	Site	Status	Last sign-in	Actions
Anupam Baruah	a.baruah@oilindia.in	HSE Analyst	Duliajan Field HQ	✓ Active	25 Sep 19:43	Change role Reset password Disable
Bikash Das	b.das@oilindia.in	HSE Analyst	OCS-2 Naharkatia	✓ Active	never	Change role Reset password Disable
D. Manikandan	d.manikandan@oilindia...	Administrator	Duliajan Field HQ	✓ Active	now	You
Kaushik Bora	k.bora@oilindia.in	HSE Analyst	Tank farm Duliajan	■ Disabled	never	Change role Reset password Enable
Priyanka Saikia	p.saikia@oilindia.in	HSE Analyst	Rig S-12 Moran	✓ Active	never	Change role Reset password Disable
Rupjyoti Gogoi	r.gogoi	Viewer	—	✓ Active	never	Change role Reset password Disable

Create HSE account

Full name

As on employee record

Email

name@oilindia.in

Role

Viewer

Site

Duliajan Field HQ

Department

HSE — Field Operations

Permissions from role

✓ View reports, evidence and dashboards

— Upload and ingest documents

— Record review decisions

— Engines, settings, logs and accounts

Permissions follow the role. To change them, change the role.

Account status

Active

Disabled until enabled here

Issue a one-time password (changed at first sign-in)

Create account...

PURPOSE

Create and manage HSE accounts: name, email, role, site, status and last sign-in.

THE PERSON

- Change role, Reset password, Disable / Enable each account
- Create HSE account: full name, email, role, site, department, status
- Issue a one-time password (changed at first sign-in); Clear form

SENTRA

- Shows the permissions that follow from the chosen role
- Marks reset requests from Forgot password
- Writes every change to the audit log

NEXT ->
Data & Backup.

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Data & Backup

Local SQL database · vector index · encrypted off-site backup

Sync now

Back up now

SQL database

18

SQLite · reports

Vector index

18

hashing

Last sync

19:44

automatic after each run

Last backup

19:44

schedule: every day

Local SQL database

Connected

SQLite · /tmp/tmpuhzv1pyy/s.db

Database URL

sqlite:///tmp/tmpuhzv1pyy/s.db

Test

Use this database

Every analysis run and decision is kept here.

Reports18

Decisions1

Action Items12

Audit entries20

☒ Sync to this database after every analysis run and review decision

Sync pushes this workstation's reports, decisions, action items and audit entries, then pulls what other workstations stored. It is keyed on content, so running it twice changes nothing.

Vector database

similar-report search

18 indexed

Encoderhashing: lexical-hash (512-d)

Reports indexed18 of 18 in this session

Not yet indexed0

Build / top up the index

10 closest report(s) by hashing: lexical-hash (512-d).

Find reports like...

scaffold board gave way, harness not clipped

Search

Ref	Similarity		Risk		IOGP rule
NM-2608	0.30		65		Driving
NM-2616	0.30		94		Working at Height
NM-2617	0.29		98		Energy Isolation
NM-2614	0.28		65		Well Control & Process Containment
NM-2602	0.28		100		Working at Height

Cloud backup

encrypted before it leaves this machine

Configured

Store backups in

Folder / synced drive

Folder

/tmp/tmpuhzv1pyy/OneDrive/SENTRA

Browse...

A folder the OneDrive, Google Drive or Dropbox client syncs is uploaded by that client; a network share is written directly.

Backups at the target

Refresh

Archive	Size	Stored
SENTRA-vm-20260925-194403.sentra	31 KB	2026-09-25 19:44:03

PURPOSE

The local SQL database, the vector index for similar-report search, sync, and encrypted off-site backup.

THE PERSON

- Database URL: Test, Use this database; sync after every run and decision
- Vector database: Build / top up the index; Find reports like...
- Cloud backup: set the target; Back up now; Sync now; restore

SENTRA

- Counts reports, decisions, action items and audit entries in the database
- Sync is keyed on content - running it twice changes nothing
- Backups are encrypted before they leave the machine (folder, S3 or WebDAV)

NEXT ->
Profile / Sign out.

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The rules SENTRA works by

When is a report SIF potential?

- A high-energy source is present (gravity / height, electrical, suspended load, pressure, fire, toxic atmosphere, traffic, ground, thermal)
- AND a critical barrier failed (fall protection, LOTO, permit / JSA, gas test, exclusion zone, bypassed safety device, fire prevention...).
- High energy with every barrier in place is controlled work - not SIF. A barrier failure without high energy is a housekeeping issue - not SIF.
- Risk score 0-95, never 100; critical from 85. The engine assesses; a person decides.

Work-hold recommendation

- Weighted evidence from: SIF precursor (+3), critical exposure (+3), high-energy hazard (+2), control failed or absent (+2), high-risk activity (+1), and the asset's memory (repeated failures, emerging precursors).
- 12 or more: Work-Hold Recommended - stop, or keep on hold, until the failed controls are restored and an HSE expert releases it.
- 4 or more: review. Below: continue - the report stays in the asset's memory.
- A reviewer's Not SIF releases a hold; every step is audited.

Security and audit

- Accounts hashed with PBKDF2; 5 failed sign-ins lock the account for 5 minutes.
- Portals check the role: an HSE account cannot sign in through Admin Login.
- Every action is written to a hash-chained audit log - an edited entry breaks the chain and the check says so.
- Secrets (database password, cloud keys) sealed in the vault, never in settings.

Data, sync and backup

- Reports, decisions, actions, audit, asset memory and work-holds in the local SQL database (SQLite by default, any SQLAlchemy URL).
- A vector index finds similar past reports.
- Backups are Fernet-encrypted archives to a folder, Amazon S3 (SigV4) or WebDAV - verified, restorable, schedulable.
- Nothing leaves the machine unless a backup target is set.